

Name:  Fahrenheit

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Objectives:

- (1.) I shall convert a measurement from a customary unit to a metric unit.
- (2.) I shall convert a measurement from a metric unit to a customary unit.

Measurement: Mass

First Unit: Customary Unit (Fluid Ounce)

To convert to: Metric Unit (Milliliter)

Second Unit: Metric Unit (Milliliter)

To convert to: Customary Unit (Fluid Ounce)

Container: Native Body Wash



First Conversion: Converting 18 fluid ounce to milliliter

From the Tables that were given:

$$1 \text{ mL} = 10^{-3} \text{ L} = 0.001 \text{ L}$$

$$1 \text{ L} = 0.26417205 \text{ gal}$$

$$1 \text{ gal} = 128 \text{ fl. oz}$$

I shall be using the Unity Fraction Method to do the conversion

$$\text{fl. oz} * \frac{\text{gal}}{\text{fl. oz}} * \frac{\text{L}}{\text{gal}} * \frac{\text{mL}}{\text{L}}$$

$$18 \text{ fl. oz} * \frac{1 \text{ gal}}{128 \text{ fl. oz}} * \frac{1 \text{ L}}{0.26417205 \text{ gal}} * \frac{1 \text{ mL}}{0.001 \text{ L}}$$

$$= 532.3235369 \text{ mL}$$

But the measurement on the container is 532 mL

This implies that the company rounded down to the nearest whole number.

Second conversion: Convert 532 milliliter to fluid ounce

$$1 \text{ mL} = 10^{-3} \text{ L} = 0.001 \text{ L}$$

$$1 \text{ L} = 0.26417205 \text{ gal}$$

$$1 \text{ gal} = 128 \text{ fl. oz}$$

I shall be using the Unity Fraction Method to do the conversion

$$\text{mL} * \frac{\text{L}}{\text{mL}} * \frac{\text{gal}}{\text{L}} * \frac{\text{fl. oz}}{\text{gal}}$$

$$532 \text{ mL} * \frac{0.001 \text{ L}}{1 \text{ mL}} * \frac{0.26417205 \text{ gal}}{1 \text{ L}} * \frac{128 \text{ fl. oz}}{1 \text{ gal}}$$

$$= 17.98905992 \text{ fl. oz}$$

Rounded answer on the container: 18 *fl. oz*

Conclusion: This implies that the company rounded up to the nearest whole number.

References: (MLA 8)

Chukwuemeka, Samuel. "Measurements and Units." *Quantitative Reasoning Project: Measurements and Units on Containers*,
conferencepresentations.appspot.com/Projects/ArithmeticAlgebra/MeasurementsUnits.html. Accessed 4 Dec. 2024.